

Water Quality C - Water Quality Div C - Rickards Invitational Div. C SATELLITE - 11-01-2025
Student Answer Sheet (/rickards/ES/AnswerSheet?tid=000GIX)

Instructions (shown before students start the test)

Hey everyone! Welcome to Water Quality Div C! You will have 50 minutes to complete this test, which assesses your understanding of freshwater ecology and water quality. Good luck!

Introduction (shown after students start the test)



1. (1.00 pts) Lentic systems are fast-moving bodies of water

☐ True ☐ False

2. (1.00 pts) What is the most defining characteristic of wetlands?

- ☐ A) Soil that is saturated with water for part of the year
- ☐ B) Oxygen poor soil as a result of constant flooding
- ☐ C) The region between bodies of water and land
- ☐ D) The presence of herbaceous vegetation that can grow in hypersaturated soils

3. (1.00 pts) What is the main difference between ponds and lakes?

- ☐ A) Ponds have a pelagic zone but lakes do not
- ☐ B) Lakes have a profundal zone where light cannot penetrate through but ponds do not
- ☐ C) Lakes are larger than ponds
- ☐ D) Ponds are created from an accumulation of rainfall, whereas lakes are created through erosion and other geologically-driven processes

4. (1.00 pts) What is an accurate characterization of phytoplankton and zooplankton?

- ☐ A) Both zooplankton and phytoplankton are autotrophs
- ☐ B) Both zooplankton and phytoplankton form symbiotic relationships with nektons in freshwater ecosystems
- ☐ C) Phytoplankton eat zooplankton
- ☐ D) Zooplankton can swim weakly; phytoplankton are generally stagnant

5. (1.00 pts) At what temperature does water have its maximum density?

- ☐ A) 0 degrees C
- ☐ B) -3 degrees C
- ☐ C) 4 degrees C
- ☐ D) 1 degree C

6. (1.00 pts) In which zone of lotic systems would you find the most producers?

- ☐ A) Source zone
- ☐ B) Transition zone
- ☐ C) Flood plain zone
- ☐ D) Rapid zone

7. (1.00 pts) What is ALWAYS the last step of the wastewater treatment process?

- ☐ A) Clarification
- ☐ B) Disinfection
- ☐ C) Bar screening
- ☐ D) Bioremediation

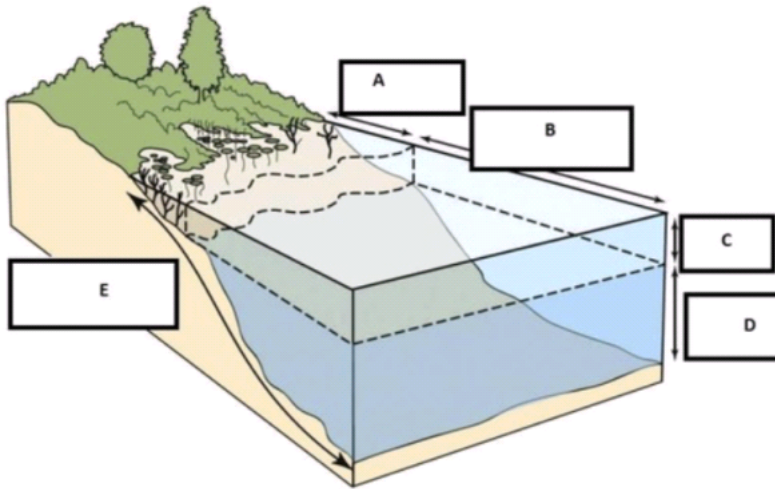
8. (1.00 pts) What is the only part of a sewage plant that is separated from the rest of the wastewater treatment facility?

- ☐ A) Bar screen
- ☐ B) Tox screen
- ☐ C) Disinfectant
- ☐ D) Clarifier

9. (1.00 pts) What property of water allows a water strider to walk across it?

- ☐ A) Adhesion
- ☐ B) Buoyancy
- ☐ C) Polarity
- ☐ D) Surface Tension

10. (5.00 pts) Label A -- E on the diagram below:

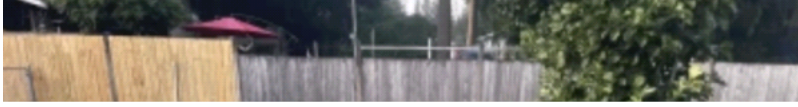


Make sure to SPECIFY what you're labeling. For example: A = green, B=blue, etc

11. (2.00 pts) Considering the distribution of plants, nutrients, and sunlight in aquatic ecosystems, why do you think large bodies of water are not highly productive?

12. (2.00 pts)

In some rural communities, where large wastewater treatment plants are not readily accessible, homes have septic tanks in their backyards, which process waste. Considering that information, explain why there's a random patch of green grass in this backyard:



13. (1.00 pts) A predator-prey relationship in which both the predator and prey are evolving is an example of which phenomenon?

- ☐ A) Niche differentiation
- ☐ B) Court Jester Hypothesis
- ☐ C) Disruptive Selection
- ☐ D) Red Queen Hypothesis

14. (1.00 pts) Which of the following statements about the Lotka-Volterra model is true?

- ☐ A) The Lotka-Volterra model only applies in situations in Hardy-Weinberg equilibrium

- ☐ B) In population cycles, the prey population lags behind the predator population
- ☐ C) In the absence of prey, predator population would decline linearly
- ☐ D) Predator population is solely dependent on prey population
- ☐ E) In the absence of prey, predator population would decline logarithmically

15. (1.00 pts) (select all that apply) Which of the following are valid solutions to the paradox of the plankton?

(Mark **ALL** correct answers)

- ☐ A) "Kill-the-winner" lytic viral hypothesis
- ☐ B) Metabolic differentiation theory
- ☐ C) Competitive Stasis
- ☐ D) Fluctuating environmental conditions
- ☐ E) Differential Predation

16. (1.00 pts)

In an aquatic environment, oysters with light-colored shells are able to camouflage in sandy areas. Similarly, oysters with dark-colored shells are favored in darker areas or in areas in shade. Camouflage enables oysters to avoid predation. What type of selective pressure would apply on a population of oysters with shells that are neither particularly light or dark?

- ☐ A) Disruptive selection
- ☐ B) Stabilizing selection
- ☐ C) Directional selection

17. (1.00 pts) Order the following by rate of nitrogen fixation from lowest to highest:

Oligotrophic lakes, Eutrophic lakes, Freshwater lakes, Saltwater oceans

- ☐ A) Oligotrophic lakes, Freshwater lake, Eutrophic lakes, Saltwater oceans
- ☐ B) Saltwater oceans, Oligotrophic Lakes, Eutrophic Lakes, Freshwater lakes
- ☐ C) Freshwater lakes, eutrophic lakes, saltwater oceans, oligotrophic lakes

- ☐ D) Saltwater oceans, Oligotrophic lakes, Freshwater lakes, eutrophic lakes

18. (3.00 pts) Describe how increased carbon dioxide levels as a result of climate change drives extinction of coral reefs. Include mention of the chemical processes involved.

19. (1.00 pts) How does the level of dissolved oxygen and carbon dioxide change with increased temperature?

- ☐ A) The level of dissolved oxygen and the level of dissolved carbon dioxide increase
- ☐ B) The level of dissolved oxygen and the level of dissolved carbon dioxide decrease
- ☐ C) The level of dissolved oxygen and the level of dissolved carbon dioxide decrease
- ☐ D) The level of dissolved oxygen decreases and the level of dissolved carbon dioxide increases

20. (1.00 pts) What is the primary source of phosphorus in aquatic environments?

- ☐ A) Atmospheric diffusion
- ☐ B) Terrestrial runoff
- ☐ C) Erosion from phosphate rocks
- ☐ D) Hydrothermal vents

21. (2.00 pts) What is a thermocline?

22. (1.00 pts) Why do some pioneer species acidify soil?

- ☐ A) To reduce toxicity of organic compounds in the soil
- ☐ B) To create salts which promote microbial activity
- ☐ C) To increase the availability of nutrients
- ☐ D) To increase the amount of insoluble compounds in the soil

23. (1.00 pts) If a person eats too much fish, they will experience mercury poisoning. What biological process does this describe?

- ☐ A) Biomagnification
- ☐ B) Bioaccumulation
- ☐ C) Bioremediation
- ☐ D) Biodegradation

24. (3.00 pts)

Amphibians that have permeable skin indicate _____ (water pollution/good oxygen levels/nutrient levels). Microalgae indicate _____ (water pollution/good oxygen levels/nutrient levels). Mayflies indicate _____ (water pollution/good oxygen levels/nutrient levels).

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25. (1.00 pts) Which of the following are impacts of harmful algal blooms?

- ☐ A) Reduced aquatic turbulence
- ☐ B) Improved water clarity
- ☐ C) Decreased dissolved oxygen levels
- ☐ D) Absorption of chemical toxins

26. (1.00 pts) How does human agricultural activity cause harmful algal blooms?

- ☐ A) Irrigation processes for crops increase the rate of harmful algal blooms
- ☐ B) Pesticides applied to crops inhibit various aquatic species and increase the risk of a harmful algal bloom

☐ C) Agricultural runoff from fertilize rich in Phosphorous and Nitrogen stimulates algal growth and causes blooms

☐ D)
Industrial burning of crops like corn as biofuel releases compounds that increase the acidity of nearby freshwater bodies, providing conditions more suitable for rapid algal growth.

27. (1.00 pts) What type of pollution does agricultural runoff represent?

☐ A) Light pollution

☐ B) Point-source pollution

☐ C) Non point-source pollution

☐ D) Organic acid pollution

28. (1.00 pts) What human activity has the largest effect on habitat fragmentation within freshwater ecosystems?

☐ A) Dam construction

☐ B) Agricultural runoff

☐ C) Industrial pollution

☐ D) Captive breeding programs

29. (1.00 pts) Agricultural runoff/chemical pollution causes algal blooms and dead zones in which many aquatic species suffocate

☐ True ☐ False

30. (1.00 pts) Bycatch leads to the rapid decline of certain species within freshwater ecosystems, and this can have larger effects on overall ecosystem health

☐ True ☐ False

31. (1.00 pts) Acid raid caused by human activities alters the chemical environment of freshwater ecosystems, endangering many species within freshwater ecosystems

☐ True ☐ False

32. (1.00 pts) Water abstraction for use in industry or human consumption leads to an increased risk of flooding

☐ True ☐ False

33. (3.00 pts) Describe 3 effects of human hydropower on freshwater ecosystems

34. (1.00 pts) How does hydropower impact water quality?

(Mark **ALL** correct answers)

- ☐ A) Causes thermal stratification
- ☐ B) Decreases dissolved oxygen levels on the top layer of a thermal stratified reservoir
- ☐ C) Definitively increases water clarity
- ☐ D) Causes fluctuating water flow which suspends biofilms and particles, increasing turbidity

35. (2.00 pts) Describe the two ways in which hydropower results in an increase in greenhouse gas emissions. Mention the two gases primarily released.

36. (2.00 pts) Explain the ways in which human use of biofuels for energy indirectly damages freshwater ecosystems

37. (2.00 pts)

Questions 37- 40 reference the picture below. This diagram is an oxygen demand curve, or a sag curve, and shows the relationship between pollutant concentration and oxygen saturation



What would the X and Y axes of this graph be labeled, respectively?

38. (2.00 pts) What is biological oxygen demand and why is it inversely related to dissolved oxygen?**39. (2.00 pts)**

As you can see from the graph, the concentration of dissolved oxygen is lowest in the section to the right of the area with the highest pollutant concentration? Why is this the case?

40. (2.00 pts) List two organisms from the rules that would be found in the third zone of this graph (lowest DO concentration):**41. (2.00 pts)** Questions 41-42 are about the organism below:



41) ID this organism with its common name

42. (2.00 pts) While this insect's larvae spends almost its entire lives feeding, adults don't eat at all because their entire purpose is to _____ (fill in the blank)

43. (2.00 pts) Questions 43-44 are about the organism below:



43) ID this organism with its common name

44. (2.00 pts) Although this organism is used as bait in aquariums, it can cause _____ disease in _____

45. (2.00 pts) Questions 45-47 are about the following organism:



45) ID this organism with its common name

46. (2.00 pts) Is the organism shown in the image male or a female? Is it the minority or majority gender in the population?

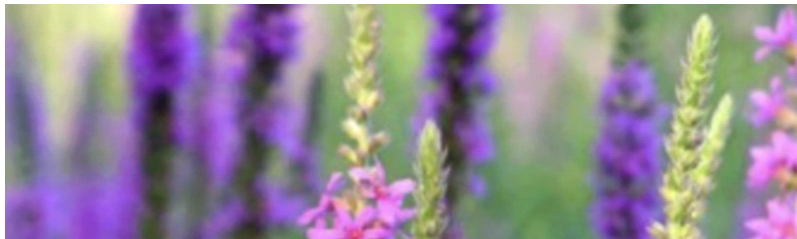
47. (2.00 pts) How does the mating behavior of the organism above support Trivers' theory of parental investment?

48. (2.00 pts) Questions 48-49 are about the following organism:

48) ID what organism this is

49. (2.00 pts) The adult form of this organism is the vector of the parasitic nematode, *Onchocerca volvulus*, which causes what disease?

50. (2.00 pts) Questions 50-51 are about the organism below:



50) ID this organism

51. (2.00 pts) This organism is invasive. Describe two general features of invasive species that allow them to outcompete native populations

52. (4.00 pts) What characterizes **nekton** and **benthos** organisms, respectively (define what they are)? Give one example of each.

53. (1.00 pts) Which of the following processes does **not** result in the formation of a lake?

- ☐ A) Volcanic activity
- ☐ B) Glaciations
- ☐ C) Crustal displacement
- ☐ D) Diagenesis

54. (2.00 pts) What is the deep-sea giantism theory? How can the phenomenon described by this theory benefit deep-sea-dwelling creatures?

55. (3.00 pts) Questions 55-57 are about the following diagram:

55) Label what type of lakes **a**, **b**, and **c** are, respectively

56. (4.00 pts)

All three diagrams have dotted lines showing the general divide between aerobic and anaerobic regions. Based on the type of lake picture **A** is, why does it make sense that it has aerobic conditions throughout? Similarly, based on the kind of lake picture **C** shows, what features of it result in it having highly anaerobic conditions?

57. (2.00 pts)

How can anaerobic conditions, similar to the ones present in picture **C**, be exploited to produce biogas? (Hint: your answer should include something about algae and lipid extraction)

58. (2.00 pts)

Eutrophication is the accumulation of excess nutrients in bodies of water that cause algal blooms. What is eutrophication, specifically caused by human activities, called?

59. (1.00 pts) What percentage of the Earth's surface is made up of freshwater systems?

- ☐ A) 1%
- ☐ B) 2%
- ☐ C) 3%
- ☐ D) 4%

60. (1.00 pts) Which zone of lotic systems has the highest concentration of dissolved oxygen?

- ☐ A) Source zone
- ☐ B) Transition zone
- ☐ C) Watershed zone
- ☐ D) Floodplain zone

61. (2.00 pts) What are two benefits that wetlands provide humans?

62. (8.00 pts) Using the table below, calculate the survivorship for each respective age:

63. (5.00 pts)

Referring to the table above (Question 62), would this species display a Type 1, Type 2, or Type 3 survivorship curve, and why? What makes each survivorship type different from the others?

64. (4.00 pts) Calculate the Shannon Wiener index for the table above. Show your work!! Round your answer to two decimal places

Animal	Population
Dragonfly	40
Crayfish	15
Frog	55
Trout	8
Heron	13
	Total: 131

65. (2.00 pts) It is often said that coral reefs have high inertia but low resilience. What does this mean in terms of succession?

66. (3.00 pts) What is the difference between point source and nonpoint source pollutants and give an example of each

67. (1.00 pts) What is the average pH of rainwater? Choose the closest answer

- ☐ A) 5.5
- ☐ B) 5.9
- ☐ C) 6.3
- ☐ D) 6.8

68. (2.00 pts) Why is rainwater acidic? List the chemical reaction that results in the acidity of rainwater.

69. (4.00 pts) How do fertilizers lead to fish populations dying in nearby lakes (explain the entire process of eutrophication for full credit)?

70. (4.00 pts)
Describe the ascorbic acid method (also known as the molybdenum blue method) and how it's used to detect phosphate concentrations in wastewater? Be sure to mention the ac created from the reaction between ammonium molybdate and phosphate and how spectrophotometers are used

71. (2.00 pts) What do the National Primary Drinking Water Regulations regulate?

72. (1.00 pts)
Which of the following is **not** true about the Surface Water Treatment Rules, a set of guidelines that outlines how surface water should be treated for human consumption?

- ☐ A) Requires most water systems to filter and disinfect water from surface water before releasing it for public consumption
- ☐ B) Establishes maximum contaminant levels for viruses, bacteria, and *Giardia lamblia*
- ☐ C) Includes treatment technique requirements

- ☐ D) Mandates that surface water must be treated at least twice to ensure pathogen safety levels

73. (2.00 pts) What are the two common methods for removing coliform bacteria from potable water?

74. (2.00 pts) What is the most common freshwater contaminant?

- ☐ A) Arsenic
- ☐ B) Lead
- ☐ C) Mercury
- ☐ D) Copper

75. (1.00 pts) What is the acceptable limit for coliform bacteria in potable water?

- ☐ A) 0 colonies/100mL
- ☐ B) 3 colonies/100mL
- ☐ C) 5 colonies/100mL
- ☐ D) 7 colonies/100mL